Lab08_Teixeira

Pedro Rocha Teixeira - 253670

```
library(reticulate)

py_install("polars")
```

```
Using virtual environment "\\smb/ra253670/Documentos/.virtualenvs/r-reticulate" ...

+ "\\smb/ra253670/Documentos/.virtualenvs/r-reticulate/Scripts/python.exe" -
m pip install --upgrade --no-user polars
```

```
import polars as pl

basics = pl.scan_csv("title.basics0.tsv.gz", separator="\t", null_values=r"\N")
ratings = pl.scan_csv("title.ratings.tsv.gz", separator="\t", null_values=r"\N")
principals = pl.scan_csv("title.principals0.tsv.gz", separator="\t", null_values=r"\N")
```

1. Quais são os 5 filmes com as maiores notas (`averageRating`)? Apresente uma solução capaz de desempatar os filmes baseando-se no número de votos recebidos.

```
res1 = (
    basics.filter(pl.col("titleType") == "movie")
    .join(ratings, on="tconst", how="inner")
    .with_columns([
        pl.col("averageRating").cast(pl.Float64),
        pl.col("numVotes").cast(pl.Int64)
    ])
    .sort(by=["averageRating", "numVotes"], descending=[True, True])
    .select(["primaryTitle", "averageRating", "numVotes"])
    .limit(5)
    .collect()
)

print(res1)
```

```
shape: (5, 3)

 primaryTitle               averageRating   numVotes
 ---                        ---             ---
 str                        f64             i64

 Kaveri                     10.0            1023
 Kurukku                    10.0            451
 Jedal Dar Omghe 30 Metri   10.0            142
 Sargashte                  10.0            134
 Gorgeous Rascal            10.0            115
```

2. Qual é o gênero mais frequente entre os filmes com nota maior que 8?

```
res2 = (
    basics.filter(pl.col("titleType") == "movie")
    .join(ratings, on="tconst", how="inner")
    .with_columns(pl.col("averageRating").cast(pl.Float64))
    .filter(pl.col("averageRating") > 8.0)
    .with_columns(pl.col("genres").str.split(","))
    .explode("genres")
    .filter(pl.col("genres").is_not_null() & (pl.col("genres") != ""))
    .group_by("genres")
    .agg(pl.len().alias("frequencia"))
    .sort("frequencia", descending=True)
    .limit(1)
    .collect()
)

print(res2)
```

```
shape: (1, 2)

 genres        frequencia
 ---           ---
 str           u32

 Documentary   10564
```

3. Quais são os 3 atores/atrizes que mais participaram de filmes com nota maior que 7.5?

```
res3 = (
    principals.filter(pl.col("category").is_in(["actor", "actress"]))
    .join(basics.filter(pl.col("titleType") == "movie"), on="tconst", how="inner")
    .join(ratings, on="tconst", how="inner")
    .with_columns(pl.col("averageRating").cast(pl.Float64))
    .filter(pl.col("averageRating") > 7.5)
    .group_by("nconst")
    .agg(pl.len().alias("num_filmes"))
    .sort("num_filmes", descending=True)
    .limit(3)
    .collect()
)

print(res3)
```

```
shape: (3, 2)

 nconst      num_filmes
 ---         ---
 str         u32

 nm0004660   231
 nm0595934   155
 nm3183374   124
```

```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-10-14 16:41:55 -03"
```